MILITARY SPECIFICATION

ADHESIVE SYSTEM, ADHESIVE AND PRIMER, FOR BONDING ETHYLENE-PROPYLENE RUBBER TO METAL

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an adhesive system for bonding ethylene-propylene rubber to metal with an adhesive and primer.

1.2 Military part number. The military part number shall consist of the letter B, the basic number of this specification, and dash designator denoting container size of primer and adhesive.

Example: B 47310 - PT

- PT — Container size (see table I)
- 47310 — Basic specification number
- B — Letter B denoting bulk material

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL, 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

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TABLE I. Container size.

Container 1/	Dash identification designator
Ounce	OZ
Pint	PT
Quart	QT
Gallon	GL

1/ Wherever inch pound units of measurement are expressed in this specification, metric equivalents shall be acceptable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/5 Aluminum Alloy, Alclad 2024, Plate and Sheet

TT-M-261 Methyl Ethyl Ketone, Technical

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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American Society For Testing And Materials (ASTM)

ASTM D429	Rubber Property - Adhesion to Rigid Substrates, Standard Test Method for
ASTM D1084	Viscosity of Adhesives, Standard Test Method for
ASTM D1875	Density of Adhesives in Fluid Form, Standard Test Method for
ASTM D3951	Commercial Packaging, Standard Practice for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified a first article sample shall meet all the requirements of this specification (see 4.4 and 6.2).

3.2 Materials.

3.2.1 Composition and conditions in container.

3.2.1.1 Composition. The adhesive compound shall be a product formulated from a polychloroprene elastomer and the primer compound shall be a product formulated from nonvolatile isocyanites, polymers, and organic compounds dispersed in an aromatic solvent (see 4.6.2.1).

3.2.1.2 Condition in container. The materials in containers shall be free from gel particles and foreign matter that would adversely affect their intended purpose (see 4.6.2.1).

3.2.2 Storage life. The primer and adhesive compounds, as furnished by the manufacturer, shall be capable of meeting the requirements specified herein

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after storage at a temperature between 16° and 32° Celsius(C) or between 60° and 90° Fahrenheit(F) in original, unopened containers for not less than six months (see 4.6.2.2).

3.2.3 Properties of uncured compounds. The properties of the compounds prior to the mixing and curing shall be as specified in table II (see 4.6.2.3, 4.6.2.4, and 4.6.2.5).

TABLE II. Properties of uncured compounds.

Property	Requirements	
	Primer	Adhesive
Solids content, percent--------	15 ± 3	30 ± 3
Viscosity, centipoises (cps)---	210 ± 50	800 ± 150
Density, pounds per gallon (lbs/gal)----------	7.4 ± 0.2	7.1 ± 0.2

3.3 Performance characteristics.

3.3.1 Peel strength. The peel strength of the adhesive-primer system shall be a minimum of eight pounds per inch width when applied and cured as specified herein (see 4.6.2.6).

3.4 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in conformance with this specification (see 4.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all

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products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection and testing of the adhesive shall be classified as:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3. Inspection conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature: Room ambient (16 to 32°C) (60 to 90°F)

b. Altitude: Facility ground

c. Humidity: Facility ambient up to 95 percent relative humidity

4.3.1 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.4 First article inspection. When specified in the contractual document (see 6.2), a sample consisting of two pints of adhesive and two pints of primer, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to first article inspection. Subsequent units will not be considered for acceptance until government approval of the first article sample has been obtained. Units subjected to first article inspection shall have successfully passed quality conformance inspection. Testing of the first article sample, to determine compliance with characteristics listed in table III, shall be conducted in accordance with the corresponding test paragraph.

TABLE III. First article inspection.

Characteristic	Requirement source	Test paragraph
Storage life-------	3.2.2	4.6.2.2
Solids content-----	table II	4.6.2.3
Density-----------	table II	4.6.2.5
Peel strength------	3.3.1	4.6.2.6

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4.4.1 Test sequence. Testing sequence for first article inspection shall be at the option of the supplier.

4.5 Quality conformance inspection.

4.5.1 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be conducted in accordance with MIL-STD-105, at an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.5.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.5.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.5.1, to determine compliance with the characteristics specified in table IV, shall be conducted in accordance with the corresponding test and inspection paragraphs.

TABLE IV. Quality conformance inspection.

Characteristic	Requirement source	Test paragraph
Composition and condition in container	3.2.1	4.6.2.1
Viscosity	table II	4.6.2.4
Peel strength	3.3.1	4.6.2.6
Workmanship	3.4	4.8
Packaging and packing	5.1	4.7
Marking	5.2	4.7

4.5.4 Test sequence. Testing sequence for quality conformance inspection shall be at the option of the supplier.

4.5.5 Noncompliance. If a sample fails to pass all inspections at the AQL, acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the contracting activity has been taken. After corrective action has been taken, all inspections shall be repeated on additional units.

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4.6 Methods of inspection.

4.6.1 Test specimen preparation. Test specimens shall be prepared of 2024-T3 aluminum (QQ-A-250/5), 0.125 inch thick, and ethylene-propylene rubber, 0.080 inch thick. The specimen configuration shall be as specified in ASTM D429. The specimens shall be cleaned and bonded as follows:

a. The rubber surface shall be solvent-cleaned and air-dried for ten minutes minimum.

b. The aluminum panels shall be cleaned in accordance with ASTM D429.

c. A single uniform brush coating of primer shall be applied to the clean dry surface of rubber and air-dried for a minimum of 30 minutes.

d. The primer on the aluminum panel shall be oven-dried at 66° ± 3°C (150° ± 5°F) for a minimum of 60 minutes.

e. The adhesive shall be uniformly applied to mating surfaces of both adherents and air-dried until the surface is totally dry to the touch; under normal conditions, 20 minutes is usually sufficient for drying. Wipe only the aluminum coated surface with clean cellulose tissues wetted with methyl-ethyl-ketone conforming to TT-M-261. Assemble the mating surfaces in 15 to 30 seconds at a minimum pressure of two pounds per square inch (psi).

f. Specimens shall be cured at 25° ± 3°C (77° ± 5°F) for a minimum of seven days for optimum strength. The pressure may be removed and specimens handled after 24 hours of curing.

4.6.2 Inspections.

4.6.2.1 Composition and condition in container. Composition shall be certified by the supplier and the condition in container shall be examined visually to establish compliance to 3.2.1.

4.6.2.2 Storage life. After storage under the conditions specified in 3.2.2, the compounds shall be subjected to all tests specified herein. The supplier's certification of compliance to the storage life requirements (see 3.2.2) may be accepted during the storage period.

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4.6.2.3 Solids content. The solids content of the adhesive and primer compounds shall be determined as follows and shall conform to table II:

a. Dry a suitable vessel in an oven at 105° ± 5°C (221° ± 10°F) and cool in a desiccator. Repeat until the weight is constant to within 0.1 milligram of last weighing.

b. Place a portion of the thoroughly mixed sample, 5.0 ± 0.2 grams into the vessel of known weight. Heat for three hours in a ventilated oven maintained at 105° ± 2°C (221° ± 4°F). Cool and weigh the dish.

c. Calculate the solids content as follows:

$$\text{Solids contents, percent} = \frac{C-A}{B} \times 100$$

Where: A = weight of vessel in grams
B = grams of sample used
C = weight of vessel and contents after heating in grams

4.6.2.4 Viscosity. The viscosity of uncured adhesive and primer compounds shall be determined at 25° ± 2°C (77° ± 3°F) in accordance with method B of ASTM D1084 except that an LVT Brookfield viscometer or equivalent shall be used, operating at 30 revolutions per minute with a number 2 spindle. The viscosity shall conform to table II.

4.6.2.5 Density. The density of uncured adhesive and primer shall be determined at 25° ± 2°C (77° ± 3°F) in accordance with ASTM D1875 and shall conform to table II.

4.6.2.6 Peel strength. The peel strength shall be determined at 25° ± 3°C (77° ± 5°F) in accordance with method B of ASTM D429 with cross-head speed of two inches per minute, except that the bond shall be adhesive in accordance with this specification and on samples prepared in accordance with 4.6.1 rather than vulcanized. The peel strength shall conform to 3.3.1.

4.7 Inspection of packaging and marking. The inspector shall ascertain that the packaging and marking of the material shall conform to the requirements of this specification.

4.8 Workmanship. The adhesive and primer compounds shall be visually examined for conformance to requirements specified in 3.4.

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5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified by the procuring activity, the packaging and packing shall be in accordance with ASTM D3951.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information (see 4.7):

a. Manufacturer's name and address.

b. Military part number.

c. Manufacturer's designation.

d. Net weight or volume.

e. Date of manufacture.

f. Shelf life or storage limitations.

g. Number of this specification.

6. NOTES

6.1 Intended use. The adhesive system is primarily intended for, but not limited to, use in bonding ethylene-propylene type synthetic rubber to metal.

6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents should specify, but not be limited to, the following information:

a. Title, number, and date of this specification.

b. Military part number.

c. Whether first article sample is required (see 3.1).

d. Requirement for certification of composition.

e. Level of preservation and packaging (see 5.1).

f. Container size and quantity.

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6.2.2 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first production lot. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Subject term (key word) listing.

Adhesive
Adhesive system
Bonding
Primer

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 8040-A129)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-47310(MI)	ADHESIVE SYSTEM, ADHESIVE AND PRIMER, FOR BONDING ETHYLENE-PROPYLENE RUBBER TO METAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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